Quantification: 701_z19c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	8
interior lacuna count	8
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1995.5	1667	1355.38	px^2
major_axis_length_px	lacuna shape	69.5414	71.5408	27.7256	px
minor_axis_length_px	lacuna shape	41.5027	36.1813	21.8773	px
aspect_ratio	lacuna shape	1.803	1.9863	0.563	unitless
eccentricity	lacuna shape	0.7351	0.8627	0.2212	unitless
solidity	lacuna shape	0.7675	0.7585	0.1025	unitless
roots_count	headline	10.375	9	4.8972	count
ring_length_r30_px	headline	431.125	396.5	185.665	px
ring_length_r60_px	reported	1048.75	1013	383.515	px
owned_length_px	ownership-dependent	2070.82	1202.92	2186.39	px
edge_count	ownership-dependent	129.125	67	136.467	count
mean_edge_length_px	ownership-dependent	15.9883	16.2114	2.3596	px
perimeter_px	normalised	311.228	284.798	200.579	px
ring_area_r30_px2	normalised	9109.62	8875	3044.04	px^2
ring_area_r60_px2	normalised	22309.4	21773.5	5301.89	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9914	1	0.0242	unitless
ring_density_r30	normalised	0.046898	0.044787	0.00990742	px^-1
ring_density_r60	normalised	0.0463718	0.0464348	0.00984425	px^-1
roots_per_100px_perimeter	normalised	3.8605	3.805	1.3188	per 100 px
ring_attached_length_r30_px	network v2	363.875	294.5	195.999	px
ring_attached_length_r60_px	network v2	740.125	559.5	465.288	px
ring_length_w_r30_px	network v2	482.923	441.57	206.646	px
ring_length_w_r60_px	network v2	1178.63	1138.96	426.667	px
sholl_crossings_r10	network v2	9.875	8.5	3.7201	count
sholl_crossings_r20	network v2	12	12	5.3984	count
sholl_crossings_r30	network v2	12.125	11	5.41	count
loop_count	loops, report only	10.75	3	15.8452	count
loop_edge_share	loops, report only	0.2128	0.2397	0.1759	share
ring_width_mean_r30_px	width	6.5061	6.4946	0.8151	px

Per field
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measure	value	units
analysed_area_px2	1.03261e+06	px^2
total_skeleton_length_px	43581	px
canalicular_length_density_per_px	0.0422046	px^-1
junction_count	1136	count
junction_density_per_px2	0.00110012	px^-2
lacuna_count	8	count

lacunae_per_px2	7.75e-06	px^-2
skeleton_component_count	709	count
mean_component_length_px	61.4683	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0475535	px^-1
field_density_without_flagged_per_px	0.0392254	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	422	px
unattached_component_count	23	count
canalicular_length_density_attached_per_px	0.0422046	px^-1
field_length_density_with_unattached_per_px	0.0426133	px^-1
loop_count	131	count
loop_edge_share	0.1986	share
edge_count	2915	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.5971	px
width_median_px	6.3246	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	22002	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	4103	100.115	82.0021	1.2209	0.5737	0.6131	-0.8887	193.446	716.814	False
2	439	24.8605	23.4655	1.0594	0.3303	0.8905	-1.2149	286.371	223.007	False
3	663	44.6399	20.5821	2.1689	0.8874	0.8511	1.5097	320.038	647.3	False
4	1055	56.5297	24.2615	2.33	0.9032	0.8843	1.0383	419.101	230.957	False
5	1893	74.136	39.3577	1.8836	0.8474	0.6949	-1.3345	688.738	268.372	False
6	1441	68.9456	33.0049	2.089	0.878	0.7238	-1.5587	747.817	126.852	False
7	3491	77.9644	65.3058	1.1938	0.5462	0.7932	-1.5218	773.022	646.082	False
8	2879	109.14	44.0418	2.4781	0.915	0.6894	-1.4893	959.511	178.398	False

Network per lacuna

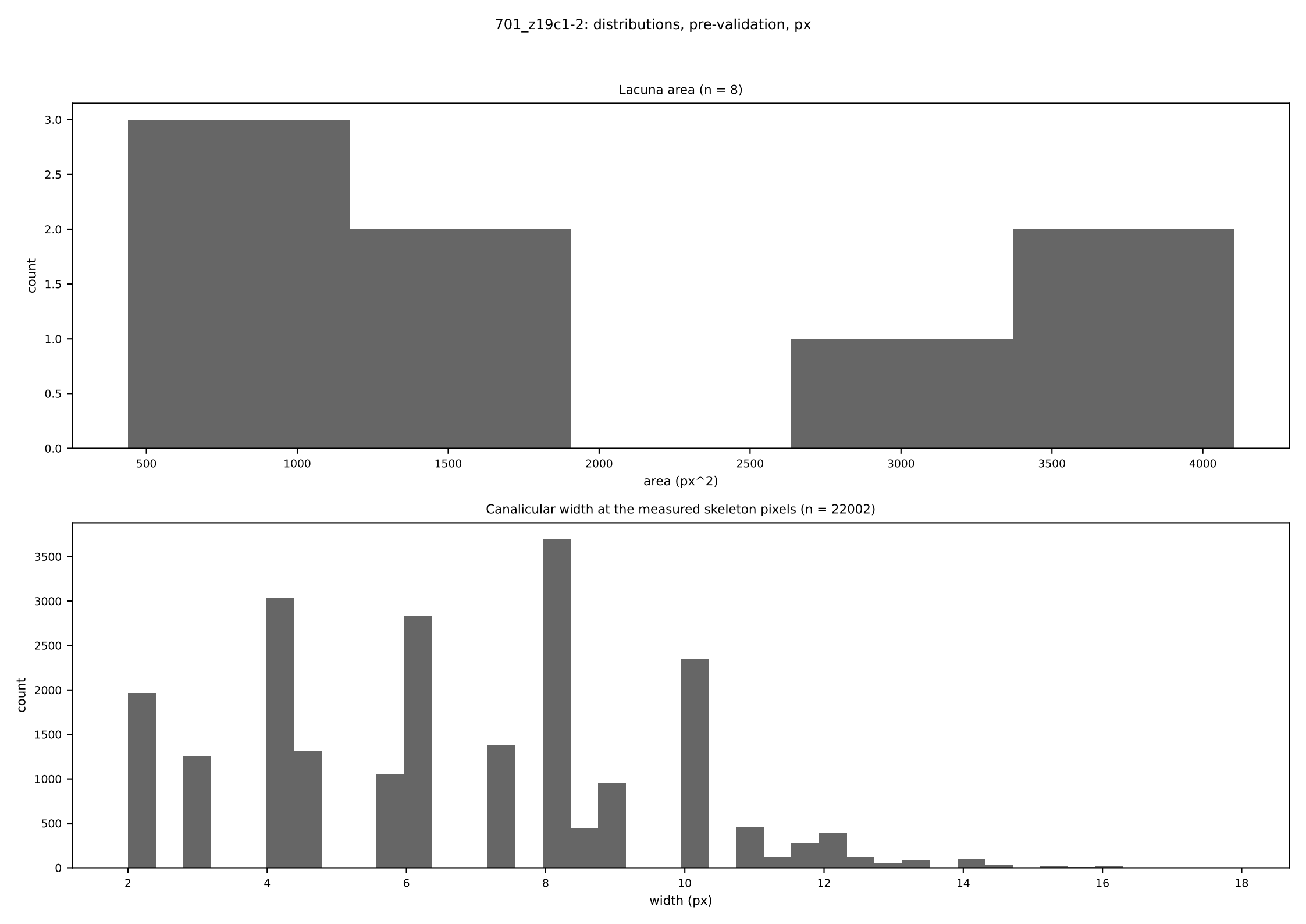
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	13	536	1121	564.902	35	16.14	5.321
2	6	276	746	931.637	52	17.9161	5.9249
3	5	226	562	253.48	23	11.0209	6.697
4	7	285	772	766.553	42	18.2513	5.9149
5	10	390	1015	2302.53	151	15.2485	6.3176
6	8	403	1011	1474.2	82	17.9781	6.6716
7	15	545	1425	6822.53	419	16.2829	7.4235
8	19	788	1738	3450.77	229	15.0689	7.7782

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	687.3	14190	30481	1	1	0.0377731	0.036777	1.8915
2	94.7107	5189	15615	1	1	0.0531894	0.0477746	6.3351
3	134.61	6194	16363	1	1	0.0364869	0.0343458	3.7144
4	162.894	6964	19128	1	1	0.0409248	0.0403597	4.2973
5	260.777	9250	22508	1	1	0.0421622	0.0450951	3.8347
6	308.818	8500	21039	1	1	0.0474118	0.0480536	2.5905
7	337.446	10732	26608	1	1	0.0507827	0.0535553	4.4452
8	503.269	11858	26733	1	0.9316	0.066453	0.0650133	3.7753

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	423	533	599.831	1257.64	13	16	9
2	191	536	312.522	852.495	6	7	9
3	195	272	250.581	627.558	7	5	6
4	242	413	328.635	875.092	7	8	8
5	256	583	434.777	1151.18	7	14	15
6	333	747	448.363	1126.75	10	10	13
7	509	1136	606.061	1588.99	13	15	14
8	762	1701	882.612	1949.38	16	21	23


701_z19c1-2: distributions, pre-validation, px
Lacuna area (n = 8)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22002)
count
width (px)